About

Data source: NASA C-MAPSS (real data)

Window: last 30 cycles
Features: 14 sensors

Alert thresholds
Warning: RUL ≤ 60 or risk ≥ 40%
Critical: RUL ≤ 30 or risk ≥ 70%

RUL is capped at 125 cycles: anything above is shown as 'healthy / far from failure'.

🛠️ Predictive Maintenance for Industrial Machinery (LSTM)

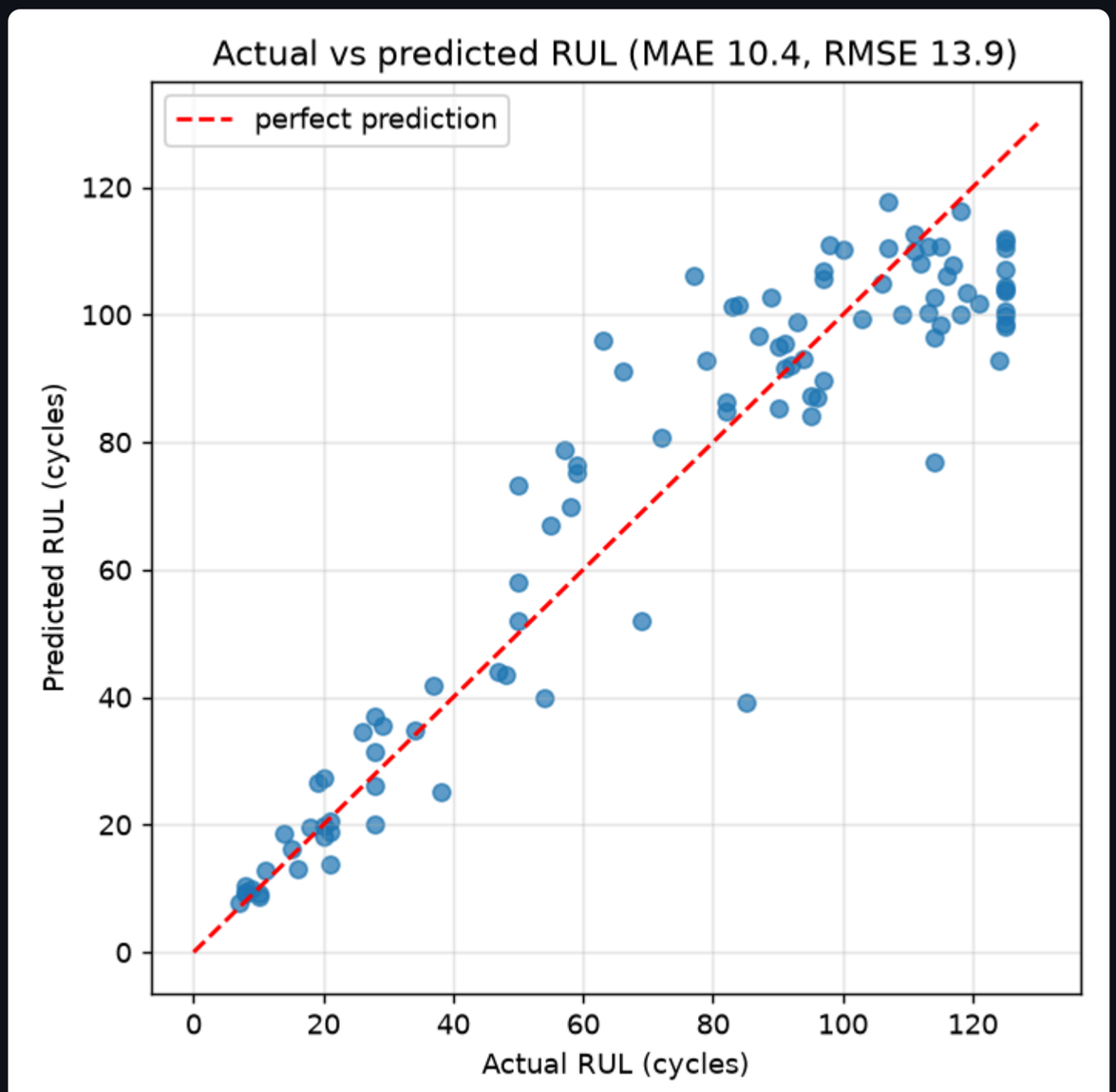
Fleet Monitor | Simulation Mode | Upload Readings | Model Performance

Computed on 100 held-out test engines · data: NASA C-MAPSS (real data)

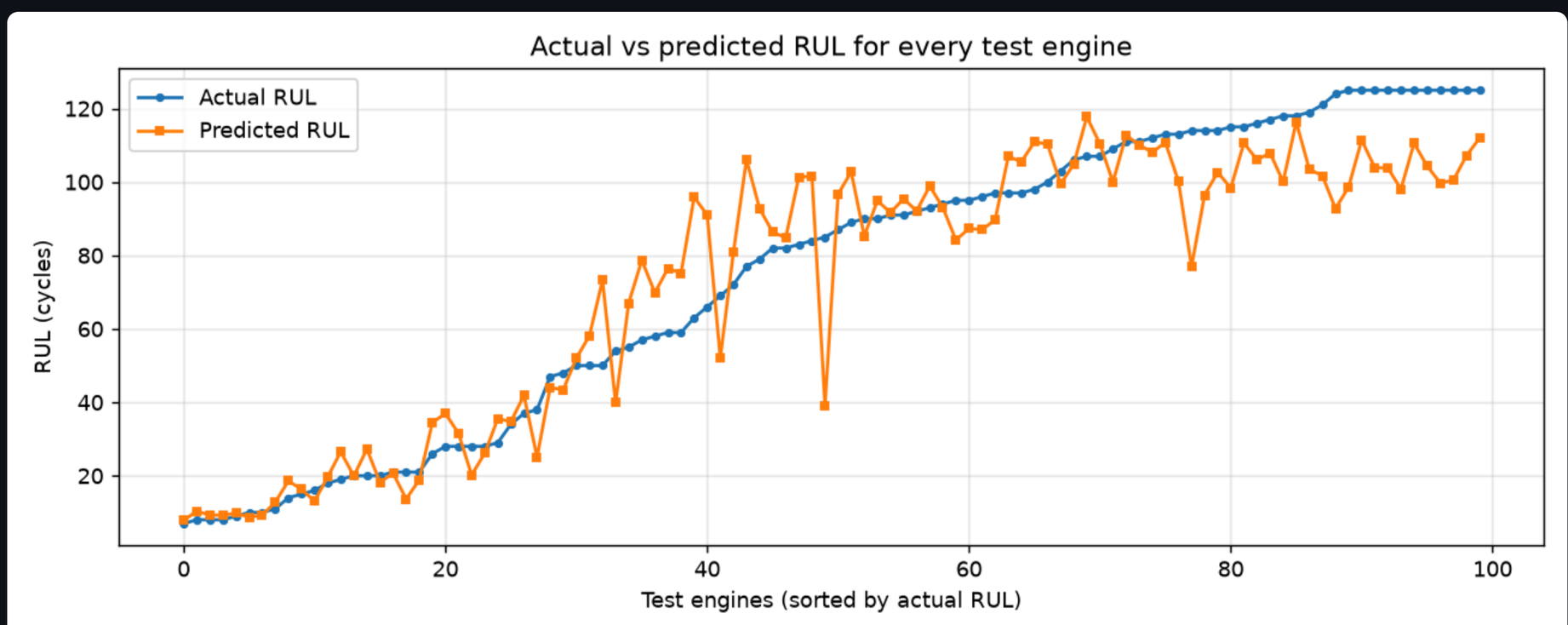
MAE (cycles)	RMSE (cycles)	R^2	NASA score
10.40	13.91	0.880	268
↑ baseline 34.83	↑ baseline 41.94		

Precision	Recall	F1-score	Accuracy
0.952	0.800	0.870	0.940

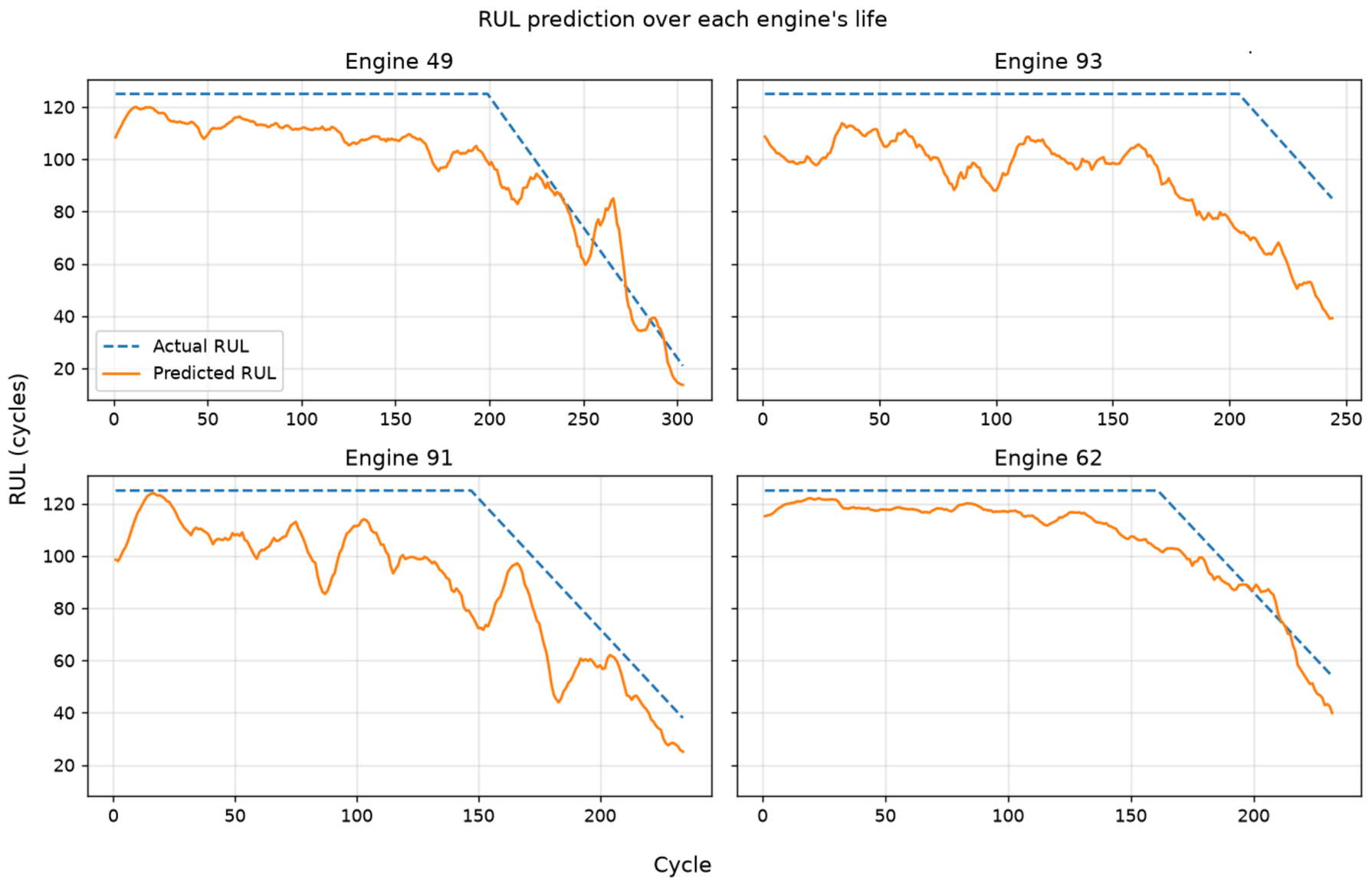
Classification = 'will fail within 30 cycles?' · 25 of 100 test engines actually were in that zone.



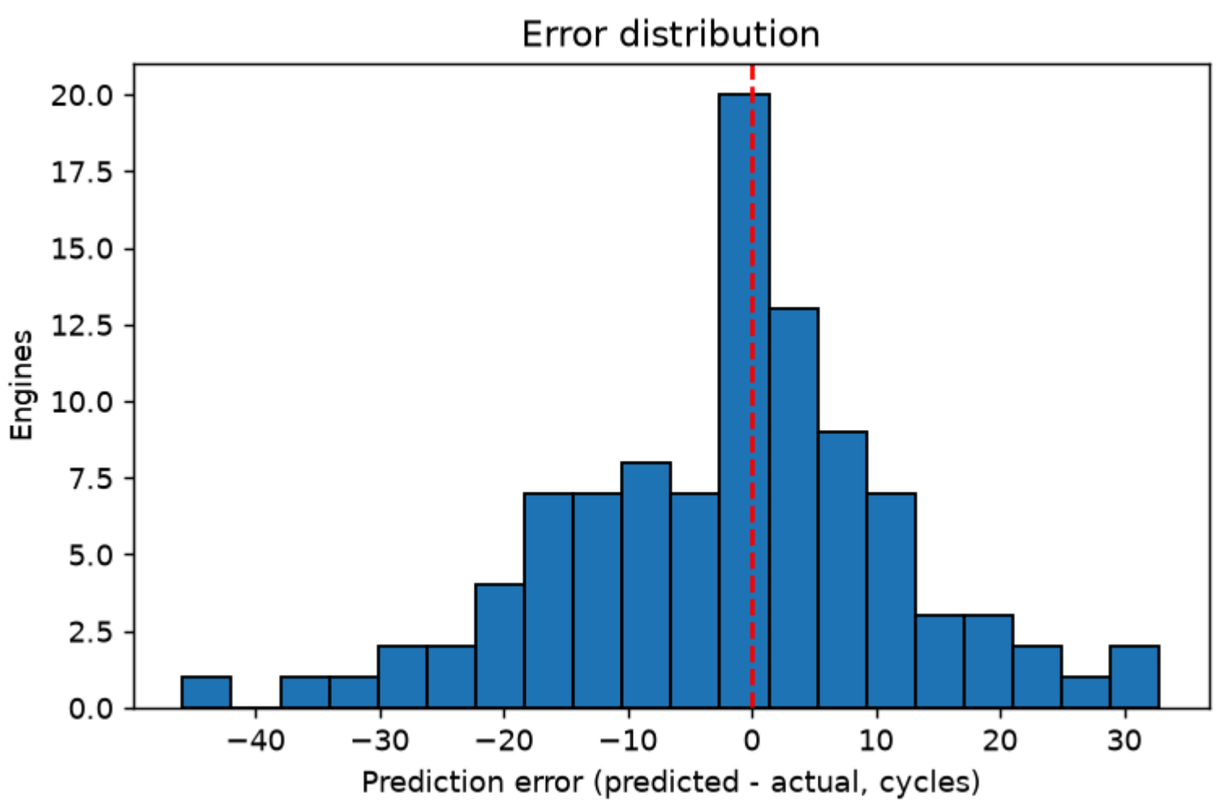
actual vs predicted scatter



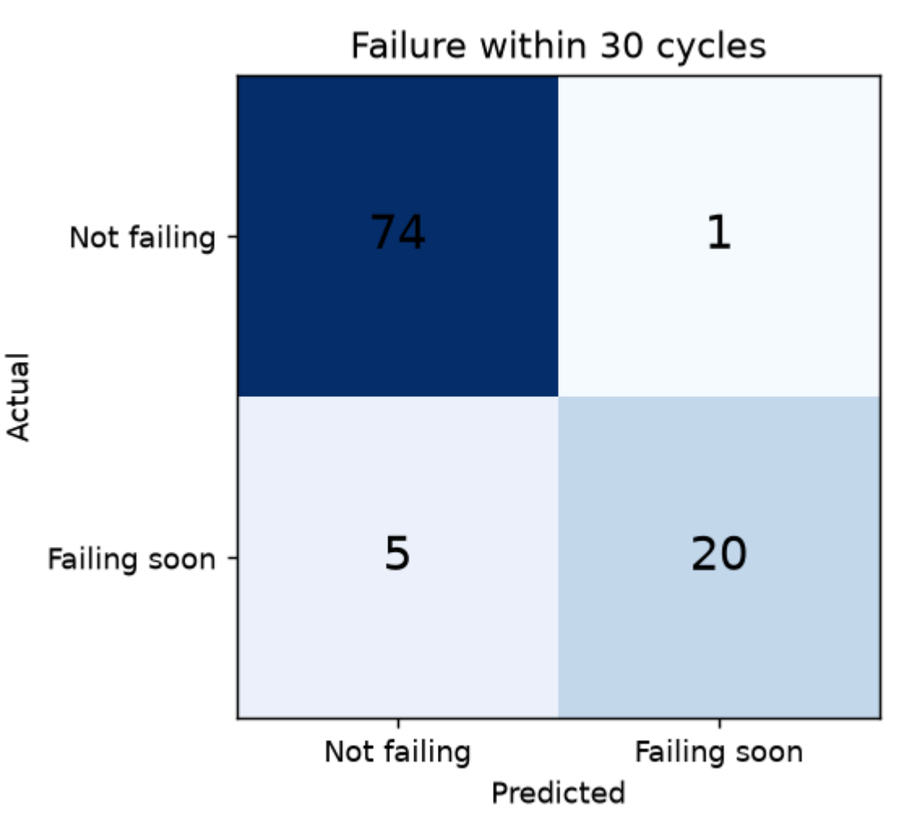
actual vs predicted by engine



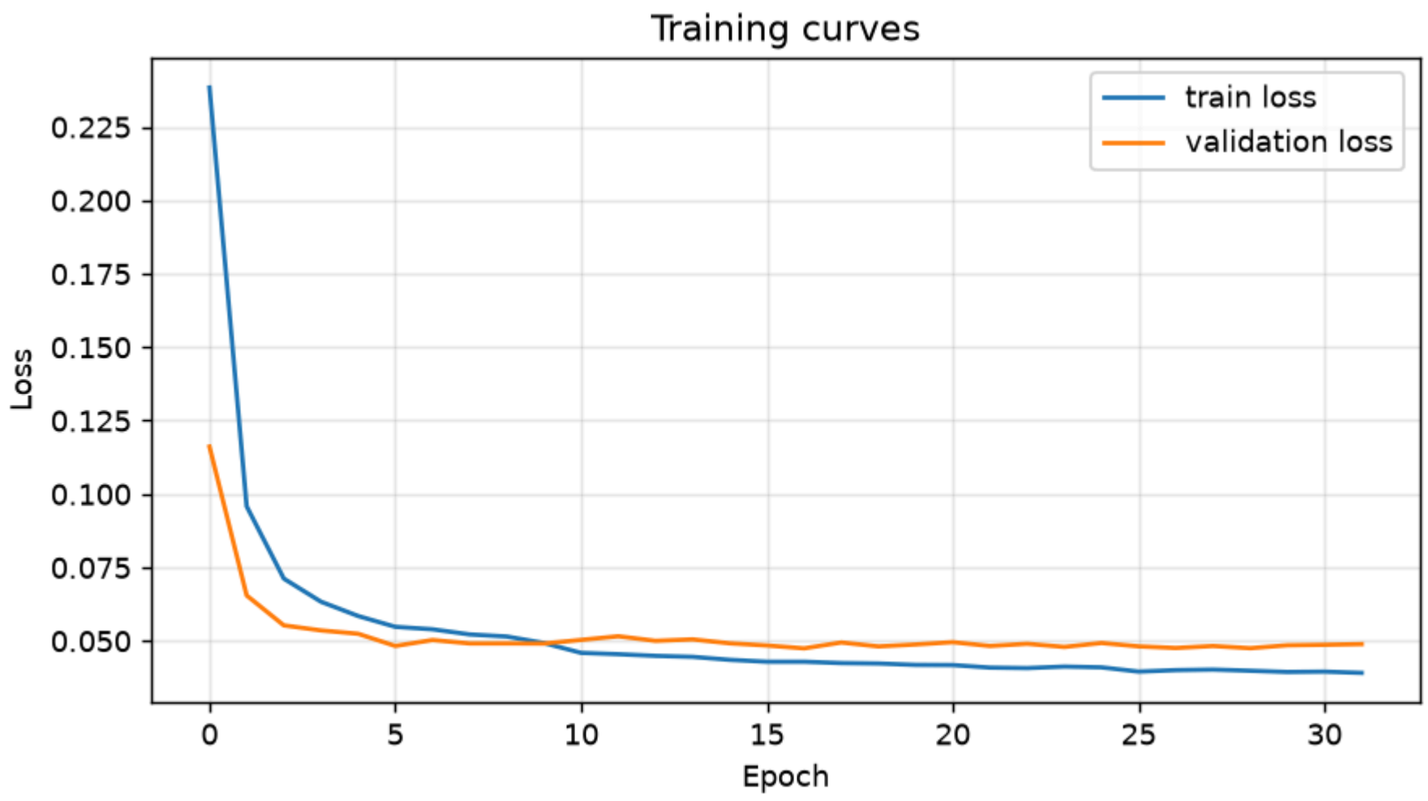
rul trajectories



error histogram



confusion matrix



training curves